MLR780 Air Conditioning and Refrigeration 2021

Assignment 5
Cooling and Dehumidifying Coils

Instructions

- This individual assignment is to be submitted online before midnight on **23 May 2021**.
- Repetitive calculation will be needed. Use a programming platform or a spread sheet of your choice. Use CoolProp for thermodynamic properties.
- Prepare a technical report that supplies all relevant equations, methods and input values, and results. Include everything that is to be graded in your report. Name it: "MLR780-2021-Assignment5-SURNAME.pdf"
- Each submission (PDF) **must** be accompanied by an official cover page obtainable from Click-UP, **as well as** your supporting documentation to prove you performed the analysis.
- The supporting documentation will not be graded, but will be checked: It must be a soft version of the script (a *.txt text file) or spread sheet with the following naming convention: "MLR780-2021-Assignment5-SURNAME.* " (The "*" represent the file extension, which will depend on the program you used – it should not be "pdf").
- **Only the PDF report will be graded.**
- Keep a time sheet documenting the time it took you to solve the questions and indicate this on the first page of your submission. Time spent on the lecture content / example / programming revision should not be included.

Outcomes:

This assignment is designed to assist you develop a basic understanding of the behaviour a cooling and dehumidification coil and how to determine the outlet conditions for a particular operating condition and temperature set-point(s).

Background:

Consider the entries of the Trane Chiller Water Fan Coil Unit catalogue on Click-UP. Our intension is to extract sufficient information from the catalogue to produce a first order simplified model of **unit 03** for an inlet water temperature ($t_{r,in}$) of **5°C**.

For this purpose, values of the thermal resistance on the air and refrigerant sides ($R_{t,a} = h_c A_a$ and $R_{t,r} = h_r A_r$ respectively) will be required, as defined by the following two equations:

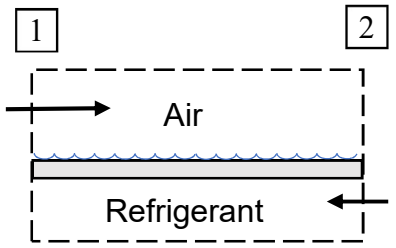
$$q_s = h_c A_a (t_a - t_i)$$
$$q_{tot} = h_r A_r (t_i - t_r)$$

where q_s is the sensible heat transfer rate on the air side, h_c is the convective heat transfer coefficient, A_a is the air side surface area, t_a is the air side dry-bulb temperature, t_i is the wetted surface temperature, q_{tot} is the total heat rate absorbed by the refrigerant, A_r is the wetted refrigerant side surface area inside the flow tube, and t_r is the refrigerant temperature.

Once these thermal resistances are known, the catalogue entries can be updated for any other inlet air and atmospheric air condition.

First show that you are able to use the equations supplied in the lecture slides by producing a model for the case where the water outlet temperature ($t_{r,out}$) is 10°C. Assume $R_{t,a}$= 300 W/K and $R_{t,r}$= 1800 W/K. Use **ONLY** the listed water and air volume flow rates and the water and air inlet states (as listed in the catalogue) and do the following:

1. Analyse the cooling coil using a single internal division (point 1 at the air inlet side and point 2 at the air outlet side) and **determine the supported sensible (q_s) and total heat transfer rates (q_{tot})** on the air side.



Show and solve the relevant equations and tabulate the converged answers for the following:

Point	**1**	**2**	**…** (for question 3)
t_a (°C)			
h_a (J/kg)			
t_i (°C)			
h_i (J/kg)			
t_r (°C)			

2. Compare the answers of question 1 with the heat rates listed in the catalogue. Comment on the validity of the assumed $R_{t,a}$ and $R_{t,r}$ values.

3. Repeat question 1, but use more than one internal division. Let N be the number of **equal surface area** divisions. For N = 2 use point index numbers 1, 2 and 3, for N = 3 use index numbers 1 to 4 etc. In each case point 1 is the air inlet side. Assume h_c and h_r to be uniform and constant. For each N that you analyse, tabulate the converged answers as before. Full equation documentation is not needed, but do document any major differences in the input values for the different N cases. Choose the number of N cases that you analyse according to what you need to answer question 4.

4. Plot the calculated q_s and q_{tot} values from question 3 against N, and comment on the impact that N has on the accuracy of your results.

For simplicity, use N = 1 for the rest of the questions. Now attempt to find the correct $R_{t,a}$ and $R_{t,r}$ by minimising the prediction errors for q_s and q_{tot} of your model.

5. Assume $R_{t,a}$= 300 W/K (not the correct value) and determine the matching $R_{t,r}$ value. Tabulate your attempts with the headings given below, and **plot** the error(s) against $R_{t,r}$.

$R_{t,r}$ [W/K]	q_s [W]	q_s error [W]	q_{tot} [W]	q_{tot} error [W]

6. Now find the correct $R_{t,a}$ and $R_{t,r}$ values (within 10%) for $t_{r,in}$ = 5°C. Use all three available catalogue entries ($t_{r,out}$ = 8, 10 and 12°C). Document your process. **Hint:** infinite number mathematical pairs of $R_{t,a}$ and $R_{t,r}$ exists for each catalogue entry, which will result in a perfect match with the catalogue. However, only one $R_{t,a}$ will also satisfy the following turbulent refrigerant side convective heat transfer relation:

$$Nu \propto Re^{0.8}$$

Remember, the size of the cooling coil remains the same for all three entries, but each entry has a different refrigerant mass rate, which influences the Reynolds number.

INDICATE HOW MUCH TIME YOU SPENT ON THIS ASSIGNMENT BY WRITING IT ON YOUR COVER PAGE